Cortical EEG slowing in resting-state Parkinson's disease

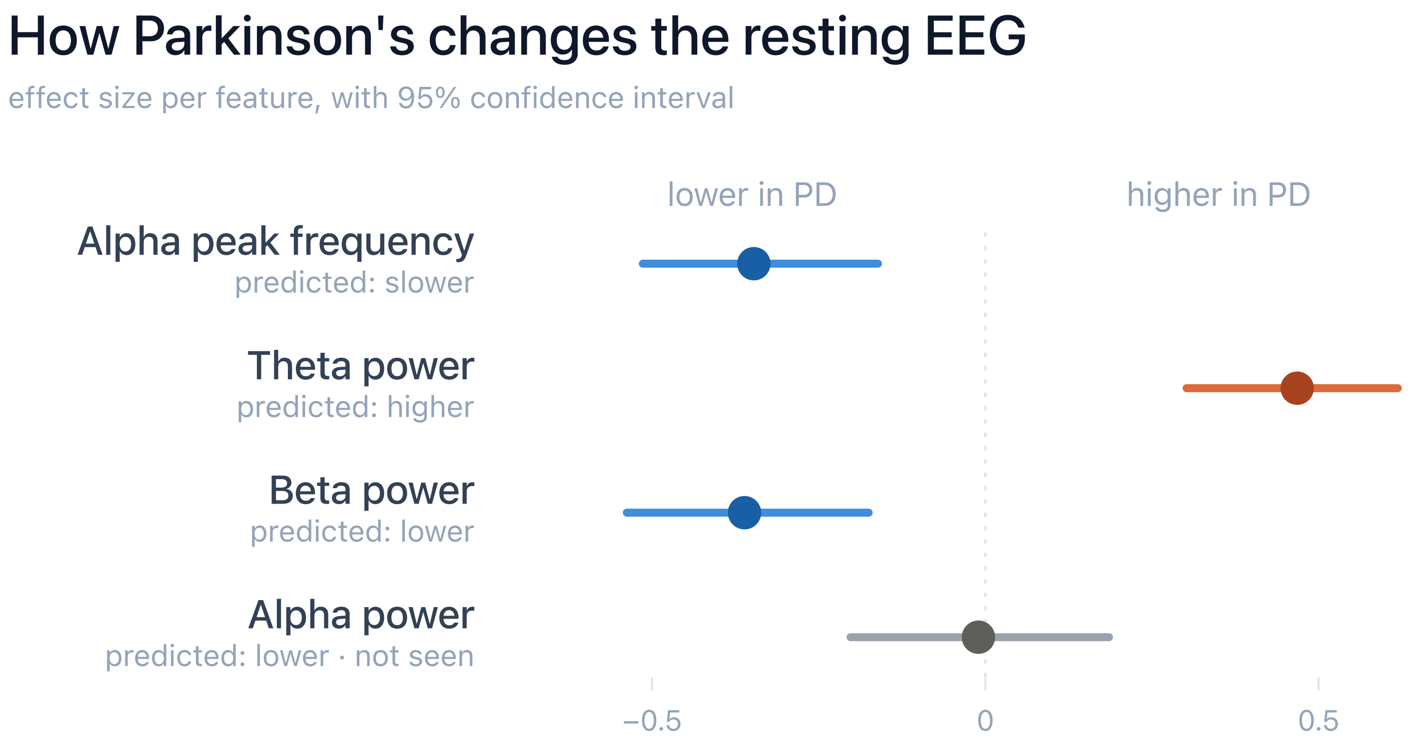
Final presentation: model, code, and findings

PD EEG group · Rui, Jan, Melissa, Friedrich · 26.06.2026

Hypotheses

Compared to age-matched healthy controls, Parkinson's patients at rest show:

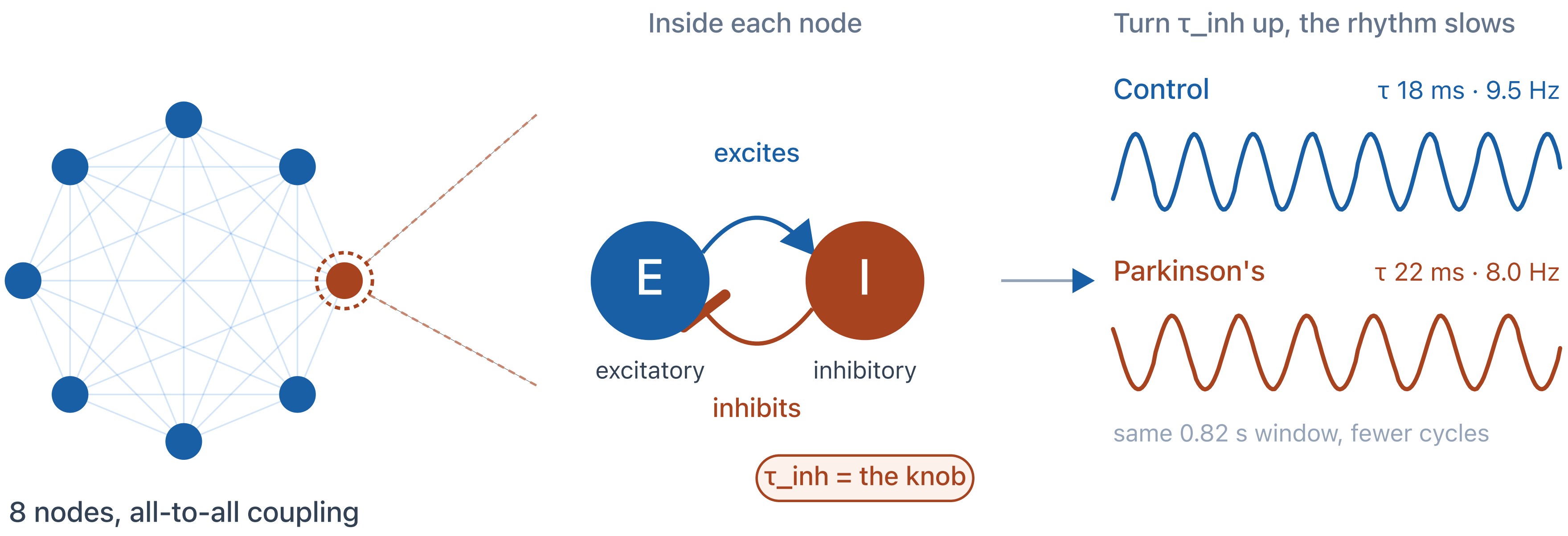
- **slowing of the alpha peak frequency** (8 to 13 Hz)
- **increased theta-band power** (4 to 8 Hz)
- **decreased cortical beta-band power** (13 to 30 Hz)



149 subjects (49 Control, 100 PD). Effect size = rank-biserial, 95% CI from bootstrap.

The model: a Wilson-Cowan network

8 coupled cortical nodes, each an excitatory-inhibitory loop. τ_inh is the only knob we turn.



The final adaptation from healthy to patient is a single parameter: a longer inhibitory time constant **τ_inh**, about **+3.25 ms** in PD.

Code & repository

THE PIPELINE

EEG to result, through six numbered scripts

one number separates the two models:
the inhibitory time constant τ_inh

EEG · 149 → **01** extract → features.csv → **03** fit → fitted_tau.csv → **05** compare → **PD vs Control**

04 data-level test · **02** builds the τ→peak curve · **06** recovers a known τ ✓

THE REPOSITORY

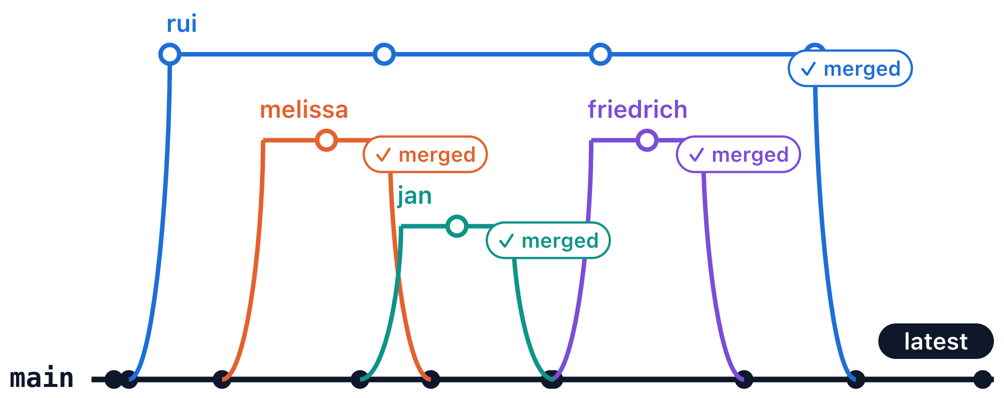
```
pd-eeg                                                     main
scripts/                                    01 … 06, numbered
  01_extract_features.py
  02_simulate_network.py
  03_fit_per_subject.py
  04_compare_features.py
  05_compare_tau.py
  06_recover_tau.py
src/                           features · model · fitting · stats
results/                                     every run writes here
  fitted_tau.csv
  tau_recovery.csv
  figures/                     compare_tau.png · tau_recovery.png
slides/                                            decks + notes
```

every result is one script run

```
bash
$ python scripts/05_compare_tau.py
Control 18.25 → PD 21.50 ms diff +3.25 p = 0.00025 ***
→ wrote results/figures/compare_tau.png
$ python scripts/06_recover_tau.py
22 known tau recovered bias −0.10 max err 1.75 ms
→ wrote results/tau_recovery.csv
```

VERSION CONTROL

feature branches, reviewed, merged into main



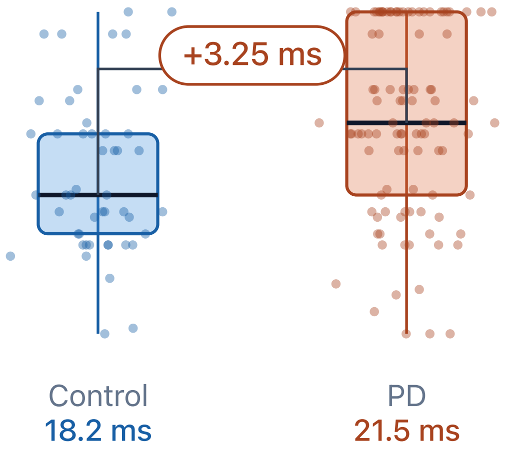
50 commits · feature branches → reviewed → merged into main · fully reproducible

Does the model separate patients from controls?

Fitted inhibitory time constant

the model parameter we change for a patient

fitting recovers tau to ±0.5 ms, far below this group gap



Validation: recovering a known parameter

simulate at a known tau, fit it back

Fitted **τ_inh** is **+3.25 ms** higher in PD (one-sided Mann-Whitney, $p \approx 0.0002$). The recovery test shows this is a real separation, not an artefact of the fitting.

Outlook and open challenges

CURRENT CHALLENGES

- One parameter (τ_inh) reproduces the alpha slowing, but not the theta and beta changes
- The fit re-expresses the data-level effect, rather than adding independent evidence

FUTURE DIRECTIONS

- Fit more than the peak (full spectrum or functional connectivity), on a real connectome
- Relate fitted τ to clinical scores and to known inhibitory (GABAergic) changes in PD

What we took away

Rui [your biggest learning]

Jan [your biggest learning]

Melissa [your biggest learning]

Friedrich [your biggest learning]

Appendix: per-subject distributions